

MASTER'S COMPREHENSIVE EXAMINATION

Part I

Theory (90 minutes)

April 10, 2010

Instructions: Attempt any **three** out of the following four questions. Mention which three you would like to have graded. Open book and open notes.

1. [Estimation] let X have the Poisson distribution with unknown mean λ .

- (a) Is 2^X a maximum likelihood estimator of 2^λ ?
- (b) Is 2^X an unbiased estimator of 2^λ ?

Prove your claim in each case.

2. [Hypothesis Testing] Let U_1, U_2, U_3 be independent draws from a population having uniform distribution on $[0, \theta]$, for an unknown θ . For testing $H_0: \theta = 10$ vs. $H_1: \theta > 10$, consider tests that reject H_0 if

$$\max(U_1, U_2, U_3) > \text{a constant.}$$

If type 1 error probability is held fixed at 25% level, what will be type 2 error probability when the true $\theta = 11$.

3. a) In a class of 30 students, 7 of the students are statistics majors, and the other 23 are not. Five students are picked at random from the class. Let A be the event that at least one of the five picked students is a statistics major. Let B be the event that at least two of the five picked students are statistics majors. Find $P(B | A)$ _____

- b) Let X be a continuous random variable with probability density function
 $f(x) = 0.4x + 0.1$, for $0 < x < 2$

Let $Y = 2X + 3$. Find the probability density function of Y .

4. a) A manufacturing process turns out items of which 0.30 are defective. If items are sampled at random, one at a time from the process, What is the probability that it takes exactly 15 items to be sampled till the 4th defective item is found _____

- b) Let X and Y be continuous random variables with joint probability density function

$$f(x, y) = \begin{cases} 21x^2y^3 & \text{for } 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal density of Y .